

Predict Heart Disease From Risk Factors

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Introduction and Motivation:

Heart disease is a common disease affecting a large population of the US. Approximately 695,000 people die from heart disease every year in the US. So, we want to study various body-related predictors in order to gauge the risk of heart disease in a certain population and the severity of their condition.

We wanted to develop a predictive model that can accurately identify whether a patient is likely to have heart disease based on various medical attributes, and how severe their condition would likely be, since early and accurate prediction can help in timely diagnosis and intervention, potentially reducing the risk of severe health outcomes

Data:

We collected our data from a public Kaggle dataset for heart disease. It was a multivariate type of dataset and was composed of 14 attributes: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, oldpeak — ST depression induced by exercise relative to rest, the slope of the peak exercise ST segment, number of major vessels and Thalassema. This dataset was used to predict a patient's risk of heart disease based on the given predictors.

Learning Formulation and Analysis:

With the health data collected, we define the learning problem to be one of classification. We will be categorizing the severity of someone's heart disease based on several predictors. We'll use logistic regression to create a model that predicts high or low impact based on the most impactful predictors. We'll split the data into a testing set and a training set, then check the accuracy of the model on the testing set. In doing so, we'll find the predictors that have the most weight. To complete our goal, we'll also use other models such as classification stacking using random forest, gradient boosting, SVC, and decision tree models. The classification stacking

model and the logistic regression model were both accurate in predicting more common severities, but had issues with predicting level 4 severity, the highest.

Experiment:

Logistic Regression: One of the methods used for the project was logistic regression. It was one of the first models created to give us the baseline of where we are starting with the data and the outcome we were working towards. During the project, we settled on the original regression as our predicted variable, num, had 5 classes indexed from 0-4. 0 indicated the patient was healthy and low risk, while 4 was severe risk. With ordinal regression, the dataset was adjusted for the imbalance of case 4, which we flagged as the most important. In our modeling, a false negative would be detrimental, so we tried to make sure these were minimized as much as possible. False positives would also occur due to the nature of modeling, but these are not as detrimental as a false negative. In medical work, a false positive would just lead to more testing, and identifying the patient in the long run is healthy or not as severe as the model had detected that patient to be. Then the model was trained on 75 percent of the data, and 25 percent was left for testing.

Random Forest: One of the other methods used for this project was a Random Forest Model. We created this because we wanted to see if we could make use of decision trees in order to predict heart disease alongside the severity of a potential patient's condition. Using 'num' as our predicted variable, we created a set of rules that would allow us to precisely cluster the data points into smaller groups so we could determine patterns in our dataset. The model was trained on 75 percent of the data, and 25 percent was left for testing.

Classification Stacking: Classification stacking compares the base model's outputs, in this case, Random Forest, Gradient Boosting, and SVC, to the true label. The model then assigns weights to the model in order to get a more accurate prediction on whether or not they have the disease. Random Forest is good against noise and is stable in predictions. Bagging trains independent models in parallel and averaging their predictions. Gradient boosting helps the model understand more subtle patterns. Boosting train models sequentially, focusing on previous errors. The support vector machine helps with the holes that these two tree methods struggle with.

Kernel-based perspective maps data to a higher dimension and generalizes the classification, working well against overfitting. Finally, logistic regression pulls them all together in order to create the overall predictions.

Decision Tree: One of the models we used in this project was a Decision Tree classifier. We selected this model because it offers clear interpretability and visual insight into how different patient features contribute to heart disease risk. We used the original *num* variable as our target, which has 5 classes ranging from 0 (no disease) to 4 (severe disease). The dataset was preprocessed by removing the *id* column and applying one-hot encoding to categorical variables to make the data suitable for training. To handle the imbalance in class distribution, specifically the underrepresentation of more severe cases, we applied the *class_weight = balanced* parameter. We limited the tree to a maximum of 3 depths and required at least 10 samples to split, which helped avoid overfitting and kept the model easy to interpret. The model was trained on 75% of the data, with 25% held out for testing.

Results

The following link has our code and graphs that we used to find our results:

https://colab.research.google.com/drive/11wPemThsjEMjfnKc8hEnRliQ0Cg9hs_H?usp=sharing#scrollTo=tQohi9ppmAsP

Logistic Regression: After predicting on the testing set using a logistic ordinal regression model, precision, recall, confusion matrix, and accuracy were used to assess the performance. The first model gave an accuracy of around 50%, and the general reading of the classes was that 0 and 1 were getting predicted well due to the larger amount of data in those classes, but the more severe cases were not being caught as much as we'd like. The next model, after fixing the data imbalance, performed better. The overall accuracy had decreased as the model was no longer biased towards the 0 and 1 classes, but there were predictions for severe cases, and the recall for class 4 had jumped from 0 to 0.4. Realistically, a 0.8 or higher is usually sought in the medical field due to the importance of correctly diagnosing those who have the disease, but for a baseline model or one of our first few models, we were making progress. The confusion matrix also gave more insight into what the model was exactly predicting and the actual correct class. Unlike the

first model, which was predicting a lot of 1's for 2's or 3's (29 cases), so always one or 2 less severe than the actual severity of the patient, this model was doing the opposite. We found the model to be biased to predicting more severe cases, which is less detrimental than the previous model. To be exact, when the patient was truly class 2 (somewhat at risk), it was more likely to predict the patient to be class 2 or above compared to the previous model. The variables used for this model are trestbps, chol, thalch, exang, oldpeak, and ca, and categorical predictors sex, dataset, cp, restecg, slope, and thal. These were the best predictors after upsampling, taking account of multicollinearity and backward regression.

Random Forest Model: After predicting on the testing set using a Random Forest Model, we decided to discard the model because we obtained an MSE of 0.534 and an R-Squared Score of 0.56, indicating that our model simply was not a good fit for our data. We then tried to use Precision and Recall to see if we could obtain better results with this model. Unfortunately, they also resulted in weak results; we obtained a Precision of 0.620 and a Recall of 0.639, indicating that the model was still not a good fit for our data. Although the accuracy was moderate and the precision was low, the model prioritized correctly identifying high-risk patients, which is critical in healthcare.

Classification Stacking: After predicting on the testing set using a Classification Stacking Model, we found that this model was a good fit as it combined multiple classifiers and balanced performance across binary classes, but not so much for several classes. For the 5-class results, the accuracy is 0.69, but F1 shows that it is 0.48, meaning the model does a really good job of identifying the healthy cases. It did very poorly at catching the severe cases (level 4), as none were predicted. In the case of just disease and no disease, where we collapse the levels so that 0 is no disease and 1-4 is has disease, we were able to get a model that performed a lot better. It had an accuracy of 0.86 and the precision and recall for both classes were above 0.8. Class 0, no disease, had a recall of 0.85 and a precision of 0.82. Class 1, which has the disease, had a recall of 0.87 and a precision of 0.89. This was another interpretation of the data, as there was nuance on what num was representing.

Decision Tree: From the classification report, the decision tree model achieved an overall accuracy of 0.50, which shows moderate performance. The model performed best on class 0 (no heart disease) with a precision of 0.83 and recall 0.57 indicating it identified healthy patients but still missed many patients. Class 1 had a recall of 0.53, showing that about half of patients with mild diseases were correctly identified. Class 2 had the highest recall of 0.66 though the precision was much lower at 0.28, which suggests that there were frequent false positives. The most concerning class was class 3, it had a recall of 0.00, which meant the model failed to correctly predict any patients with that level of disease. Class 4 had a high recall of 0.80 but really low support, making it statistically unstable. These metrics demonstrate the model's strong bias towards detecting lower severity or healthy cases while struggling with higher disease levels.

Challenges:

We initially planned to do tuberculosis and also incorporate image data into our project, but we found several issues with the attempt. The tuberculosis dataset was very poorly organized, and the number of "not reported" was the majority of the samples. Furthermore, the data was awkwardly split due to the website only allowing 5 variables/features to be selected for every download. It would lead to an enormous clean-up that may not be even usable due to the sheer amount of NA, "not reported", and imbalance of data. So we decided to find another dataset that would be more reliable.

With logistic regression, there were a lot of issues that occurred mainly due to not knowing how important a balanced dataset is. This led us to try several different methods to change the outcome of our models without actually knowing why it wasn't accurately predicting cases 4's or 3's too well. We tried to do feature correlation and remove all the highly correlated features, and also did backwards selection and dropped the p-values that were above the $\alpha = 0.05$. In hindsight, the issue was our dataset itself, but after fixing up previous thresholds and trying other suggested methods of improving the model and noticing that changes to the model didn't seem to help, we realized maybe there was another issue. This landed us at upsampling, which would later make us realize how important a balanced dataset is.

Discussion

Our logistic regression (baseline) captured minor heart disease well, but failed to detect severe cases (class 4 had 0 recall). Multicollinearity and weak predictors were removed to improve performance. Logistic Regression that used upsampling addressed class imbalance by increasing samples for rare cases. This improved recall for class 44 from 0.00 to 0.40, which is a large jump for a model that was performing recall at 0 previously. Decision Tree achieved a high recall (0.80) for class 4, catching the most severe cases. Its accuracy was moderate, and precision was low; the model prioritized correctly identifying high-risk patients, which is critical in healthcare. It also provides interpretability, showing which patient features contribute most to risk. Finally, the stacking model combined multiple classifiers and balanced performance across all classes, with improved generalization and reduced class-specific bias. Overall, the severe heart disease/risk (class 4) was hardest to predict, but recall improved significantly with model tuning. In healthcare, recall is more important than precision because it is safer to flag too many than to miss real cases. Techniques like upsampling and class weighting are essential for handling imbalanced medical data.

Updated Timeline:

Data cleaning	Learning formulation	Training and implementation	Evaluation and analysis
February 15 - March 21	February 15 - March 21	March 22 - 31	April 1 - 7

References:

Abdullah, S. M. (2024). Heart Disease Dataset | Cleaned. Kaggle.com.
<https://www.kaggle.com/datasets/abdmental01/heart-disease-dataset/data>